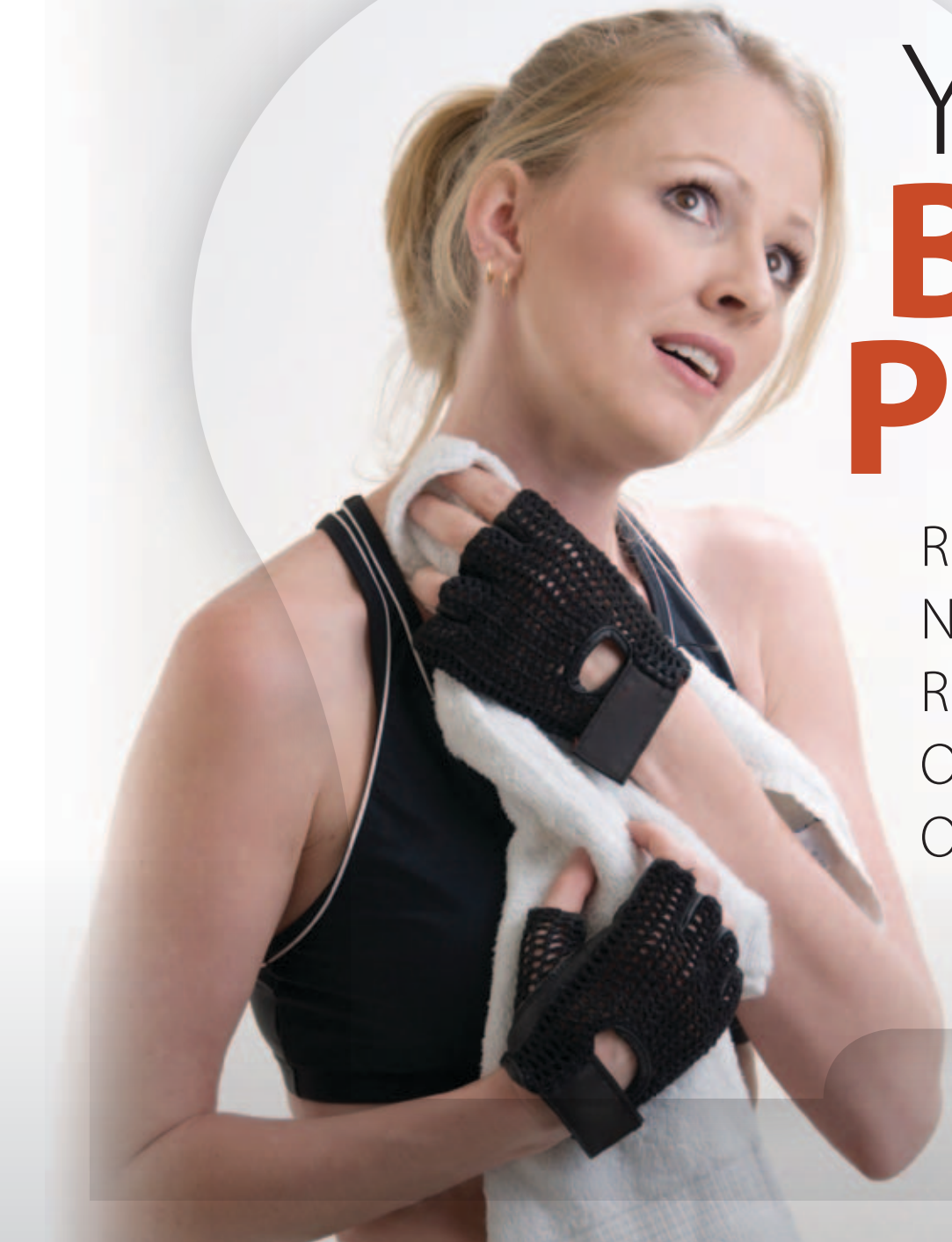
A Publication of the Wellness Council of America

WELCOA's News & Views

Maximizing Your Brain Power



Revolutionary New Science Reveals Benefits Of Exercise On Brain Function

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About John J. Ratey, MD



Dr. John Ratey is an Associate Clinical Professor of Psychiatry at Harvard Medical School, a Research Synthesizer, Speaker, Author, and Clinical Psychiatrist maintaining a private practice in Cambridge, Massachusetts. He has lectured and published 60 peer-reviewed articles on the topics of Aggression, Autism, ADHD, and other issues in neuropsychiatry, and has authored or co-authored seven books.

His latest book, *Spark: The Revolutionary New Science of Exercise and the Brain*, presents an understanding of neurobiology with worldwide research to inspire the reader to embrace exercise as a means to achieve optimal brain performance.

WELCOA's News & Views

An Expert Interview with

John J. Ratey, MD

About David Hunnicutt



Since his arrival at WELCOA in 1995, David Hunnicutt has interviewed hundreds of the most influential business and health leaders in America. Known for his ability to make complex issues easier to understand, David has a proven track record of asking the right questions and getting straight answers. As a result of his efforts, David's expert interviews have been widely-published and read by workplace wellness practitioners across the country.

David Hunnicutt can be reached at dhunnicutt@welcoa.org.

In this exclusive WELCOA interview, Dr. David Hunnicutt asks Dr. Ratey to share some of the key concepts from his book, *Spark*. The result is a fascinating discussion about the benefits of exercise and the important impact it has on our brain function. Dr. Ratey shares insights about how the latest science can help individuals and communities embrace physical activity—and in turn, help people enhance their brain function and improve their overall health.

David Hunnicutt: How important is exercise to human function?

John Ratey: I think we all know that exercise works in the body from the neck down and is instrumental in addressing and treating the chronic illnesses and conditions we face today. However, my work revolves around how exercise affects the brain. I like to say that exercise is really something we do for the brain, and the rest of the body benefits from the side effects.

DH: Can exercise make a person smarter?

JR: Yes, it can. There are several studies indicating that exercise can actually increase one's IQ. There is also significant evidence that exercise can make one more acute and can increase motivation, problem-solving skills and memory. Exercise also helps brain cells function better than anything else that we know.

> "Exercise is really something we do for the brain, and the rest of the body benefits from the side effects."

Exercise provides the right kind of "soup" our brain cells require. It enriches the environment cells need to reach their full potential. It helps boost a natural occurrence in the brain called neurogenesis—which means creating new brain cells. We didn't know until about 10 years ago that the human brain can actually make new brain cells. Every day, we can either promote or retard cell growth. Stress retards the growth of new brain cells. Learning improves it. Love improves it. Having fun improves it. And exercise really improves the number of new, functioning brain cells that can be spawned every day.

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Exercise Improves Brain-Power!

Exercise is the single most powerful tool you have to optimize your brain function. Visit the *Sparking Life* website to find out how transforming America's sedentary lifestyle could help us improve our children's learning capacity, reduce negative effects of stress, manage mental health issues, and maintain our cognitive abilities as we age.

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Exercise increases the connection between existing cells, and it also increases the actual number of cells. As a result, anyone who exercises is much more likely to be alert and motivated. At the same time, that person is less prone to distractions, fidgetiness and stress.

> "Anyone who exercises is much more likely to be alert and motivated. At the same time, that person is less prone to distractions, fidgetiness and stress."

DH: Can exercise help slow the aging process?



JR: Yes, I think most of our initial findings and excitement comes from looking at the elderly, looking at the aging process and at the prevention of the onset of Alzheimer's disease and cognitive decline. For the past 15 years, most of the neural science in this area has been dominated by the effect of exercise and its ability to stave off cognitive decline. We have found that exercise can keep the brain performing at its highest levels. In fact, there are numerous studies all over the world that validate exercise as an anti-aging tool. The literature shows that exercise can help keep elderly individuals mentally sharp. Moreover, some studies show that many areas of an elderly and/or middle-aged brain can actually grow in response to exercise. The hippocampus—the memory central station in our brain—responds particularly well to physical activity. This area is the most affected by the birth of new brain cells. So, as we're making new brain cells as a result of exercise, it is specifically affecting this area of the brain, which in turn can hopefully help us remember more. At least that's the thought behind all of this.

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DH: How important is physical activity for school kids?

JR: It's vital. We need to leverage exercise in classrooms because it can help restart the arousal system—it can help children be more attentive and less disruptive. The findings on this are remarkable. One study conducted this year in Ontario looked at a school's 25 worst performing students. These children were enrolled in a very intensive exercise program. One of the key findings after the study concluded was in regards to suspension days. Before the program, suspension days among the 25 students totaled 95 for a three-month period. After the program, it dropped to five days in a three-month period.

> "When exercise is incorporated into a child's daily routine, we see that discipline and attendance problems are reduced almost immediately."

And we see this happening frequently when exercise is incorporated into a child's daily routine. We see that discipline and attendance problems are reduced almost immediately. And it's not because physical activity tires the kids out; it's because it turns on parts of the brain that help regulate their emotional systems and their impulsivity.



WELCOA'S dyk DID YOU KNOW...

SPARK Reveals Revolutionary New Science

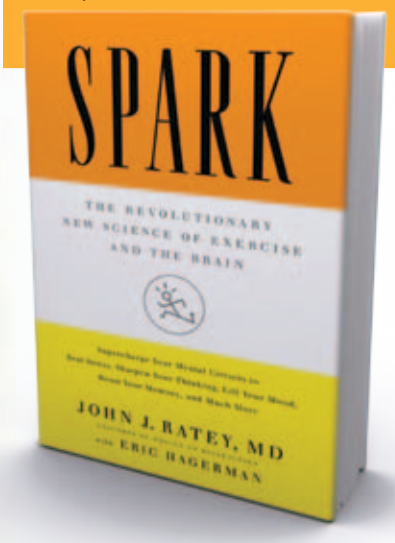
The crucial connection between exercise and the brain's performance is revealed in Dr. John Ratey's groundbreaking book, *SPARK: The Revolutionary New Science Of Exercise And The Brain*. *SPARK* will help you understand how exercise positively affects these areas of cognitive function:

- **Memory & Thinking**
- **Stress**
- **Anxiety & Mood**
- **Aging**
- **Addiction**
- **And More!**

Download an overview of this science at:

www.welcoa.org/uploads/pdf/interviews/spark-flyer.pdf

Order your copies of *SPARK* today!



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DH: How much exercise do people need to reap the kind of benefits you're talking about?

JR: I think we need to focus on at least 30 to 40 minutes of exercise a day. And in my opinion, exercise is not just walking the block or walking around the mall; it's really pushing yourself and really working at it. The good thing is that exercise can be in the guise of doing something fun, like playing a game, doing a sport or competing in an activity you enjoy. So, if you can focus on 30 to 40 minutes of moderate-intensity exercise for five to six days a week, I think you're going to find a lot of benefits.

> "Exercise is not just walking the block or walking around the mall; it's really pushing yourself and really working at it."

DH: How long will it take for people to start reaping those benefits?

JR: They'll see changes almost immediately. That first week you may be tired—especially if you've been sedentary for a while. But even after that second week, you will begin to feel more energized and focused and less stressed and anxious.



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DH: If exercise is so important, why aren't doctors prescribing it?

JR: Well, part of it is that it seems too simple. The medical profession that I grew up in often looks into one cause and one treatment for a given problem. They make a diagnosis and apply the medication. It's part of our modern culture to expect medical science to take care of our health and our chronic conditions.

However, five years ago, the President of the American College of Sports Medicine advised that every doctor, no matter what their specialty, should ask about their patients' exercise regimen and should urge them to set up an exercise program. This was echoed at the inauguration speech of the new president of the American Medical Association. So, doctors and specialists are definitely aware of the need to prescribe exercise. Of course, it's hard to change the culture, and it really is about changing the culture. It's hard for many doctors to preach and prescribe exercise if they themselves are overweight and not the ideal picture of wellness. It's hard for them to say do as I say, not as I do.

DH: What was your epiphany in terms of being more physically active? How did it happen for you?

JR: Well, I was always an athlete, but I spent my graduate years in Boston where it seemed everyone was running marathons. So, I began to really pay attention to the effect of exercise on marathoners and myself.

I got really excited about exercise when I learned about a school system that created an exercise program over the course of 20 years or so. They implemented a fitness-based exercise program in their PE program that was second to none. There were 19,000 kids in this school system in Naperville, Illinois, and only three percent of them were overweight. This school system's students also took the International Science and Math Test. Ninety-nine percent of the kids participated and they scored number one in the world in science and number six in math.

This is essentially what got me to start my research and to write my book. My own regimen changed a bit when I delved into the science and read over a thousand papers on exercise and its effect on the brain. I came upon an interval training issue, which indicated that doing short, quick, very intense bouts of exercise in the midst of short recovery times can lead to very profound changes in the body and the brain. So, I began to work that into my exercise routine.

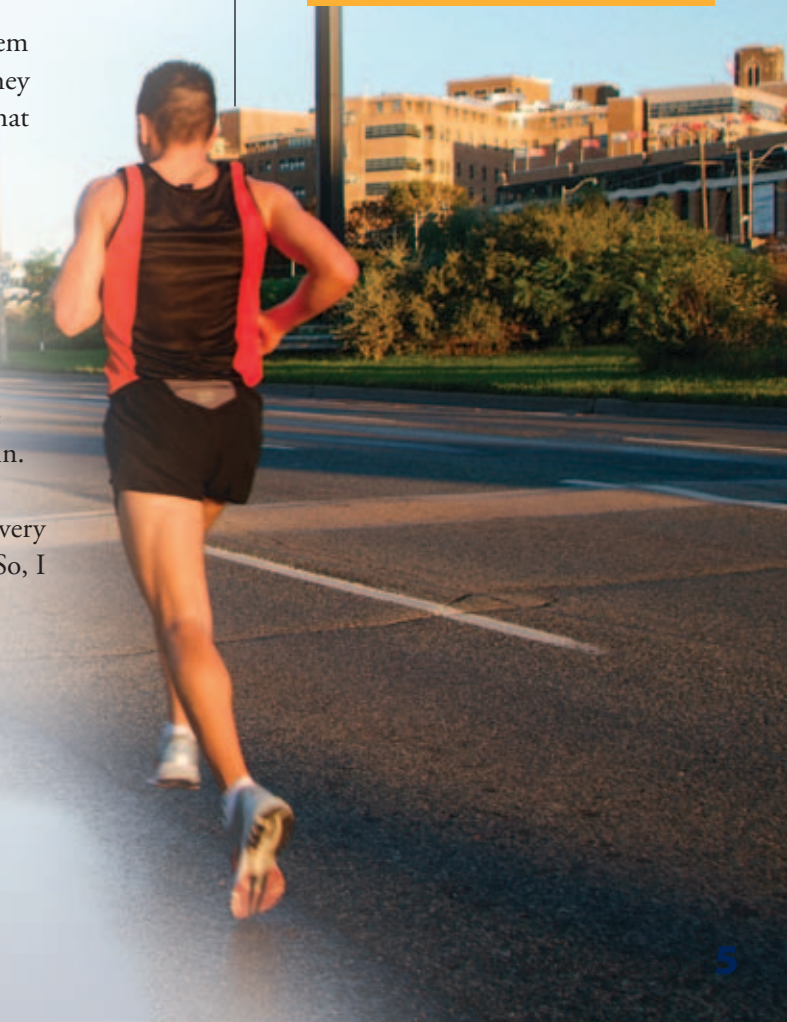
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AT-A-GLANCE SUMMARY

Notable & Quotable

from John J. Ratey, MD

Exercise Can Make You Smarter…

There are several studies indicating that exercise can actually increase one's IQ. There is also significant evidence that exercise can make one more acute and can increase motivation, problem-solving skills and memory. Exercise also helps brain cells function better than anything else that we know.

PAGE 1

Slowing The Aging Process…

For the past 15 years, most of the neural science in this area has been dominated by the effect of exercise and its ability to stave off cognitive decline. We have found that exercise can keep the brain performing at its highest levels. In fact, there are numerous studies all over the world that validate exercise as an anti-aging tool.

PAGE 2

Kids Need Exercise Too…

When exercise is incorporated into a child's daily routine…we see that discipline and attendance problems are reduced almost immediately. And it's not because physical activity tires the kids out; it's because it turns on parts of the brain that help regulate their emotional systems and their impulsivity.

PAGE 3

Feeling The Benefits Of Exercise…

[You'll] see changes almost immediately. That first week you may be tired—especially if you've been sedentary for a while. But even after that second week, you will begin to feel more energized and focused and less stressed and anxious.

PAGE 4

What We're Up Against…

We've created this [cyber] world where no one has to move much and we've become non-movers. We're still operating with the genes we had 10,000 years ago when we were hunter-gatherers moving 10 to14 miles every day. We still have that basic gene profile and we should be moving.

PAGE 7

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DH: Are you optimistic about the future in terms of increasing levels of physical activity?

JR: I'm always optimistic because I think the more science we can throw at people, the more they'll realize that this is something they need to do for their own health and happiness. Hopefully that will help stem the tide.

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> I think the more science we can throw at people, the more they'll realize that this is something they need to do for their own health and happiness.

I think we're fighting against the cyber world, which puts us in our chairs, and this is the major problem we have to face; we have to deal with this. We've created this world where no one has to move much and we've become non-movers. We're still operating with the genes we had 10,000 years ago when we were hunter-gatherers moving 10 to14 miles every day. We still have that basic gene profile and we should be moving. Of course, it's hard to get people to move 10,000 steps a day, let alone six or seven miles a day. And that's part of the reason why we have such a health care crisis. It's too bad that our focus hasn't been on a health crisis as opposed to the health care crisis. We really need to get everybody paying attention to what they can do to manage their own health.

My mission is to help schools discover the importance of exercise and to help get every child active. We need every teacher—not just PE teachers or health professionals—to know that the more fit a child is, the better the child will perform. And again, there is science from around the world to back this up. For example, Sweden had an epidemiologic study looking at 1.2 million boys over a 26-year period. They tested the participants' aerobic capacity along with their IQ and other cognitive measures. They tested the boys when they began the study and retested the same 1.2 million boys at the age of 18 when they all entered compulsory military service. They found that the boys who improved their aerobic capacity increased their IQ scores and their cognitive performance. Those who didn't improve their aerobic capacity didn't show improvements in IQ and cognitive performance. I've also done a paper recently with Dr. Jim Loehr from the Human Performance Institute on how exercise improves cognition in working-aged people. So, there's big time evidence that we have and we see it all over the place. N&V



ABOUT **Madeline Jahn**

Madeline Jahn is the Communications Associate for the Wellness Council of America. Her role is to coordinate the development of new publications and pool existing resources for WELCOA members, serving the mission of health promotion through marketing and planning support. She has a Master's Degree in Organizational Leadership from the College of Saint Mary. For questions about this publication, or to obtain permission for reprinting, please contact Maddy at *mjahn@welcoa.org*.

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